Summary

Vertebrates circulate blood in their body, to transport essential substances to the cells and to carry waste substances from them. Blood is carried away from the heart, passes through tissues in capillaries and is returned to the heart in veins. Blood pressure drops gradually as it passes along this system. Arteries have thick, elastic walls which allow them to withstand high blood pressure. Arterioles are small arteries that help to reduce blood pressure and control the amount of blood flow to different tissues. Capillaries are only just wide enough to allow the passage of red blood cells, and have very thin wall to allow efficient and rapid transfer of materials between blood and cells. Veins have thinner walls than arteries and possess valves that allow blood to flow back to the heart even at low pressure.

Blood consist of plasma and formed elements. Blood plasma leaks from capillaries to form tissue fluid. This is collected into lymphatics as lymph, and returned to the blood in the subclavian veins. Tissue fluid and lymph are almost identical in composition. They contain fewer plasma protein molecules than blood plasma as these proteins are too large to pass through the pores in the capillary walls. The formed elements of blood constitute RBC, WBC and Platelets.

The mammalian heart has four chambers, right and left artia and right and left ventricles. The separation of chambers in the heart results in complete double circulation. The cardiac cycle is a continuous process but can be considered in five stages. Beating of the heart is initiated by the sinoatrial node (SAN) or pacemaker which has its own myogenic rhythm. Blood pressure is the force exerted by blood on the walls of blood vessels, and it is responsible for moving blood through the vessels.

Cardiovascular disease accounts for more deaths each year in the India. Cardiovascular conditions include systemic hypertension, atherosclerosis, coronary artery disease, angina pectoris, myocardial infarction and stroke. Cardiovascular diagnostic techniques and treatments include cardiac angiography, balloon angioplasty, and coronary artery bypass. The circulatory system contributes to homeostasis by transporting O_2, CO_2, wastes, electrolytes, and hormones from one part of the body to another.

Evaluation

1. What is the function of lymph?
   a. Transport of O_2 into brain
   b. Transport of CO_2 into lungs
   c. Bring interstitial fluid in blood
   d. Bring RBC and WBC in lymph node

2. Which one of the following plasma proteins is involved in the coagulation of blood?
   a. Globulin **b. Fibrinogen**
   c. Albumin d. Serum amylase

3. Which of the following is not involved in blood clotting?
   a. Fibrin b. Calcium
   c. Platelets **d. Bilirubin**

4. Lymph is colourless because
   a. WBC are absent
   b. WBC are present
   c. Haemoglobin is absent
   d. RBC are absent

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